

Threshold Entry in a Fixed-Prize Contest

Anurag Srivastava

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1 Introduction

Table 1: Headline claims, each traced to the model rows it rests on.

ID	Headline	Model rows	Scope note
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2 Model

Table 2: Results of the model, each with its statement and the Lean theorems that check it.

ID	Claim	Statement	Lean	Proof
M1	Representation (P1).	$\Delta(m) = V \int \phi dH_m = V \mathbb{E}[\phi(M_m)]$, with $M_m \sim H_m$.	EntryContestP1P2.two_max_sum, EntryContestP1P2.p1_representation, EntryContestP1P2.Delta_eq_expectation	p. 4
M2	Dominance (P2).	For $s \geq 0$ and $\mu \leq 1$, $F \leq G$ and $\Delta(m) \geq 0$. For $\mu < 1$ and $\Pr(s > 0) > 0$, $F < G$ on some $[q, u)$ with $q < u$.	EntryContestP1P2.p2_fosd, EntryContestP1P2.p2_strict_interval, EntryContestP1P2.Delta_nonneg_from_primitives	p. 4
M3	Step identity (P3).	For $m+2 \leq Q$, $\Delta(m+1) - \Delta(m) = -\frac{V}{2} \int \phi^2 dK_m$.	EntryContestStep.three_max_sum_real, EntryContestStep.step_identity, EntryContestModel.Delta_step	p. 4
M4	Monotonicity.	$\Delta(m+1) \leq \Delta(m)$ for $V \geq 0$, with $<$ when $V > 0$ and $\int \phi^2 dK_m > 0$.	EntryContestModel.Delta_step_nonpos, EntryContestModel.DeltaUpTo_step, EntryContestEq.Delta_step_strict	p. 5
M5	Threshold entry (P5).	For $\kappa_0 \leq \dots \leq \kappa_{Q-1}$ and Δ non-increasing, $\kappa_j \leq \Delta(j)$ and $i \leq j$ give $\kappa_i \leq \Delta(i)$.	EntryContest.single_crossing, EntryContestModel.p5_threshold_from_primitives	p. 5

ID	Claim	Statement	Lean	Proof
M6	Count invariance.	Every equilibrium S , with $\kappa_i \leq \Delta(S - 1)$ for $i \in S$ and $\Delta(S) < \kappa_j$ for $j \notin S$, has $ S = k^*$.	EntryContestEq.count_invariance_fin, EntryContestMM.exists_mem_ge, EntryContestMM.exists_not_mem_le	p. 5
M7	Identities pinned.	If $\Delta(k^* - 1) < \kappa_{k^*}$, the assortative set $\{0, \dots, k^* - 1\}$ is the only equilibrium.	EntryContest.members_below_kstar, EntryContestEq.assortative_unique	p. 5
M8	Assortative selection (N4).	$\{0, \dots, k^* - 1\}$ is an equilibrium; $\sum_{i < k} \kappa_i \leq \sum_{i \in S} \kappa_i$ for $ S = k$; with identical score laws, payoff sums of equal-size equilibria differ by their cost sums.	EntryContestEq.assortative_is_equilibrium, EntryContestEq.assortative_min_cost, EntryContestEq.prize_term_anonymous, EntryContestEq.payoff_gap_anonymous	p. 5
M9	Benchmark (P6).	With the incumbent investing and $s \geq 0$ atomless, $\Delta(m) = V/(m+2)$ at $\mu = 0$ for every Q , and $\Delta(m) \rightarrow V/(m+2)$ as $\mu \rightarrow 0$.	EntryContestP6P7.integral_iid, EntryContestP6P7.integral_cdf_pow, EntryContestP6P7.p6_evaluation, EntryContestP6P7.p6_at_zero, EntryContestP6P7.tendsto_cdf_investorLaw, EntryContestP6P7.tendsto_cdf_nonInvestorLaw, EntryContestP6P7.p6_limit	p. 6
M10	Saturation (P7).	$\Delta(m) \leq V/(m+1)$ for every Q , and an entrant at index m has $(m+1)\kappa \leq V$.	EntryContestP6P7.integral_cdf_pow, EntryContestP6P7.p7_Delta_le, EntryContestP6P7.p7_cap, EntryContestP6P7.p7_count	p. 6
M11	Prize and fee (P8).	k^* is non-decreasing in V and, for u non-decreasing, non-increasing in c .	EntryContestEq.kstar_mono, EntryContestEq.Delta_mono_prize, EntryContestMM.Delta_scale, EntryContestEq.kappa_mono_fee	p. 6
M12	Margin condition.	For sorted profiles w and w' , $w_{k^*-1} \leq w'_{k^*-1}$ gives $k^* \leq k'^*$, and $w'_{k^*} \leq w_{k^*}$ gives $k'^* \leq k^*$.	EntryContestSpread.enters_down, EntryContestSpread.count_ge_of_enters, EntryContestSpread.count_le_of_fails, EntryContestSpread.margin_rise, EntryContestSpread.margin_fall	p. 7

ID	Claim	Statement	Lean	Proof
M13	Pivot spreads (P9-gen, P9).	For a pivot-spread T about x_0 , k^* weakly rises when $w_{k^*-1} \geq x_0$ and weakly falls when $w_{k^*-1} \leq x_0$. The linear spread about the mean is one, and keeps the mean.	EntryContestSpread. pivot_rise, EntryContestSpread. pivot_fall, EntryContestSpread. spread_isPivot, EntryContestMM.mm2a	p. 7
M14	The band.	With $d = w' - w$, the signs of d leave the direction of k^* open only when $d_{k^*-1} < 0 < d_{k^*}$. A pivot-spread never does so, and inside the band both directions occur for every small $\mu > 0$.	EntryContestSpread. branch_of_not_band, EntryContestSpread. not_band_of_single_crossing, EntryContestSpread. pivot_single_crossing, EntryContestSpread. witness_counts, EntryContestSpread. witness_signs, EntryContestSpread. band_both_directions, EntryContestSpread. band_both_directions_positive	p. 7
M15	Strictness under a spread (P9).	If some entrant lies below x_0 at $\lambda = 1$ and $\kappa(w) \rightarrow \infty$ as $w \downarrow c$, $k^*(\lambda)$ is non-increasing in $\lambda \geq 1$ and $k^*(\lambda) < k^*(1)$ above some $\hat{\lambda}$.	EntryContestKappa. exit_below_pivot, EntryContestKappa. enters_downward, EntryContestKappa.p9_strict, EntryContestKappa.p9_monotone, EntryContestKappa.p9_strict_model	p. 8

3 Literature

Table 3: Papers the model or the conclusions depend on, each with a verbatim quote.

ID	Paper	What we rely on	Quote	Page	Model rows
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4 Conclusions

Table 4: The novel claims, each linked to the model and literature rows it rests on.

ID	Statement	Rests on
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A Proofs

Notation. Challenger i scores $\sigma_i = \mu r_i + (1 - \mu)s_i e_i$, with r_i and $s_i \geq 0$ drawn independently of wealth. F and G are the score CDFs of an entrant and of an outsider, and C is the incumbent's, each without atoms. With m other entrants among the Q challengers,

$$H_m = C F^m G^{Q-1-m}, \quad K_m = C F^m G^{Q-2-m}, \quad \varphi = G - F,$$

$$\Delta(m) = V \left(\int H_m dF - \int H_m dG \right), \quad \kappa(w) = u(w) - u(w - c).$$

Challengers are indexed $0, \dots, Q-1$ in increasing order of cost, $\kappa_0 \leq \dots \leq \kappa_{Q-1}$, which is decreasing order of wealth. Index j enters when $\kappa_j \leq \Delta(j)$, and k^* satisfies $\kappa_j \leq \Delta(j)$ for $j < k^*$ and $\Delta(k^*) < \kappa_{k^*}$ when $k^* < Q$. In the results on wealth profiles, rank j has wealth w_j with $w_0 \geq w_1 \geq \dots$, and enters when $w_j > c$ and $\kappa(w_j) \leq \Delta(j)$, so a challenger whose wealth is at or below the fee does not enter. A prime marks the profile after a spread and its count k'^* .

Proof of M1

Lean: EntryContestP1P2.two_max_sum, EntryContestP1P2.p1_representation, EntryContestP1P2.Delta_eq_expectation.

For independent $X \sim A$ and $Y \sim B$ with B atomless, $\Pr(X = Y) = 0$, so

$$\int B dA + \int A dB = \Pr(Y \leq X) + \Pr(X \leq Y) = 1.$$

With $(A, B) = (F, H_m)$ and $(A, B) = (G, H_m)$,

$$\int H_m dF = 1 - \int F dH_m, \quad \int H_m dG = 1 - \int G dH_m,$$

so

$$\frac{\Delta(m)}{V} = \int (G - F) dH_m = \int \phi dH_m = \mathbb{E}[\phi(M_m)].$$

□

Proof of M2

Lean: EntryContestP1P2.p2_fosd, EntryContestP1P2.p2_strict_interval, EntryContestP1P2.Delta_nonneg_from_primitives.

Let $X = \mu r + (1 - \mu)s$ and $Y = \mu r$. For $\mu \leq 1$ and $s \geq 0$, $X - Y = (1 - \mu)s \geq 0$, so $\{X \leq x\} \subseteq \{Y \leq x\} \cup \{s < 0\}$ and

$$F(x) \leq G(x) + \Pr(s < 0) = G(x), \quad \Delta(m) = V \int \phi dH_m \geq 0.$$

For $\mu < 1$,

$$\{s > 0\} \subseteq \{Y < X\} = \bigcup_{q \in \mathbb{Q}} \{Y \leq q < X\},$$

so $\Pr(s > 0) > 0$ gives $q \in \mathbb{Q}$ with $\Pr(Y \leq q < X) > 0$, and

$$G(q) \geq \Pr(X \leq q) + \Pr(Y \leq q < X) > F(q).$$

$G - F$ is right-continuous, so $G - F > 0$ on some $[q, u)$.

□

Proof of M3

Lean: EntryContestStep.three_max_sum_real, EntryContestStep.step_identity, EntryContestModel.Delta_step.

Write $K = K_m$, so $H_{m+1} = KF$ and $H_m = KG$. For independent atomless draws $Z \sim K$, $X \sim A$, $Y \sim B$, exactly one is the largest,

$$\int AB dK + \int KB dA + \int KA dB = 1.$$

With $(A, B) = (F, F)$, (G, G) and (F, G) ,

$$\int KF dF = \frac{1}{2} \left(1 - \int F^2 dK \right), \quad \int KG dG = \frac{1}{2} \left(1 - \int G^2 dK \right), \quad \int KG dF + \int KF dG = 1 - \int FG dK,$$

so

$$\frac{\Delta(m+1) - \Delta(m)}{V} = \int KF dF - \int KF dG - \int KG dF + \int KG dG = -\frac{1}{2} \int (F - G)^2 dK.$$

□

Proof of M4

Lean: EntryContestModel.Delta_step_nonpos, EntryContestModel.DeltaUpTo_step, EntryContestEq.Delta_step_strict.

By M3 and $\phi^2 \geq 0$,

$$\Delta(m+1) - \Delta(m) = -\frac{V}{2} \int \phi^2 dK_m \leq 0 \quad (V \geq 0), \quad < 0 \quad (V > 0, \int \phi^2 dK_m > 0).$$

For $m \geq Q - 1$ the gain is extended as $\Delta(Q - 1)$, so $\Delta(m+1) \leq \Delta(m)$ for every m .

□

Proof of M5

Lean: EntryContest.single_crossing, EntryContestModel.p5_threshold_from_primitives.

For $i \leq j$, by M4,

$$\kappa_i \leq \kappa_j \leq \Delta(j) \leq \Delta(i).$$

□

Proof of M6

Lean: EntryContestEq.count_invariance_fin, EntryContestMM.exists_mem_ge, EntryContestMM.exists_not_mem_le.

Let $k = |S|$. A k -set of indices has an element $i \geq k - 1$, since $S \subseteq \{0, \dots, k - 2\}$ gives $k \leq k - 1$, and omits some $j \leq k$, since $\{0, \dots, k\} \subseteq S$ gives $k + 1 \leq k$. If $k > k^*$, then $k^* < Q$ and

$$\Delta(k^*) < \kappa_{k^*} \leq \kappa_{k-1} \leq \kappa_i \leq \Delta(k - 1) \leq \Delta(k^*).$$

If $k < k^*$, then $j < Q$ and

$$\Delta(k) < \kappa_j \leq \kappa_k \leq \Delta(k).$$

Both are contradictions, so $k = k^*$.

□

Proof of M7

Lean: EntryContest.members_below_kstar, EntryContestEq.assortative_unique.

A member i of an equilibrium has $\kappa_i \leq \Delta(k^* - 1)$ by M6. If $i \geq k^*$,

$$\kappa_i \geq \kappa_{k^*} > \Delta(k^* - 1),$$

so $i < k^*$. Then $S \subseteq \{0, \dots, k^* - 1\}$ and $|S| = k^*$ give $S = \{0, \dots, k^* - 1\}$.

□

Proof of M8

Lean: EntryContestEq.assortative_is_equilibrium, EntryContestEq.assortative_min_cost, EntryContestEq.prize_term_anonymous, EntryContestEq.payoff_gap_anonymous.

(i) For $i < k^*$ and for $k^* \leq j < Q$,

$$\kappa_i \leq \kappa_{k^*-1} \leq \Delta(k^* - 1), \quad \Delta(k^*) < \kappa_{k^*} \leq \kappa_j.$$

(ii) Induct on S , adding its largest element a last. With $T = S \setminus \{a\}$ and $T \subseteq \{0, \dots, a - 1\}$, $|T| \leq a$, so

$$\sum_{i < |T|+1} \kappa_i = \sum_{i < |T|} \kappa_i + \kappa_{|T|} \leq \sum_{i \in T} \kappa_i + \kappa_a = \sum_{i \in S} \kappa_i.$$

(iii) With identical laws, the win probability of i is $P^{\text{in}}(|S|) = \int CF^{|S|-1} G^{Q-|S|} dF$ for $i \in S$ and $P^{\text{out}}(|S|) = \int CF^{|S|} G^{Q-1-|S|} dG$ for $i \notin S$, so with $EU_i(S) = u(w_i) - \kappa_i \mathbf{1}\{i \in S\} + VP_i(S)$,

$$\sum_i EU_i(S) = \sum_i u(w_i) - \sum_{i \in S} \kappa_i + V(|S|P^{\text{in}}(|S|) + (Q - |S|)P^{\text{out}}(|S|)),$$

and the last term is equal for $|S| = |T|$. □

Proof of M9

Lean: EntryContestP6P7.integral_iid, EntryContestP6P7.integral_cdf_pow, EntryContestP6P7.p6_evaluation, EntryContestP6P7.p6_at_zero, EntryContestP6P7.tendsto_cdf_investorLaw, EntryContestP6P7.tendsto_cdf_nonInvestorLaw, EntryContestP6P7.p6_limit.

For $X_0, \dots, X_k$ independent with atomless CDF F ,

$$\int F^k dF = \Pr\left(X_0 = \max_{i \leq k} X_i\right) = \frac{1}{k+1}.$$

At $\mu = 0$ the entrant scores s and the outsider 0, so $G = \mathbf{1}_{[0, \infty)}$ and $F(0) = \Pr(s \leq 0) = 0$. With $C = F$,

$$\int H_m dF = \int F^{m+1} dF = \frac{1}{m+2}, \quad \int H_m dG = H_m(0) = 0.$$

For $\mu \rightarrow 0$ let $X_\mu = \mu r + (1 - \mu)s$ and $Y_\mu = \mu r$. For $s > 0$,

$$F_\mu(X_\mu) \rightarrow F_0(s), \quad G_\mu(X_\mu) \rightarrow 1, \quad F_\mu(Y_\mu) \rightarrow F_0(0) = 0,$$

with every integrand in $[0, 1]$, so by dominated convergence

$$\int H_m dF_\mu \rightarrow \int F_0^{m+1} dF_0 = \frac{1}{m+2}, \quad 0 \leq \int H_m dG_\mu \leq \mathbb{E}[F_\mu(Y_\mu)] \rightarrow 0.$$

□

Proof of M10

Lean: EntryContestP6P7.integral_cdf_pow, EntryContestP6P7.p7_Delta_le, EntryContestP6P7.p7_cap, EntryContestP6P7.p7_count.

From $C \leq 1$, $G \leq 1$ and $\int H_m dG \geq 0$,

$$\Delta(m) \leq V \int H_m dF \leq V \int F^m dF = \frac{V}{m+1}.$$

An entrant at index m has $\kappa \leq \Delta(m)$, so

$$(m+1)\kappa \leq (m+1)\Delta(m) \leq V, \quad m+1 \leq \frac{V}{\kappa} \quad (\kappa > 0).$$

□

Proof of M11

Lean: EntryContestEq.kstar_mono, EntryContestEq.Delta_mono_prize, EntryContestMM.Delta_scale, EntryContestEq.kappa_mono_fee.

Let k^* and k'^* be the counts for (κ, Δ) and (κ', Δ') , with $\kappa'_j \leq \kappa_j$ and $\Delta(j) \leq \Delta'(j)$ for every j . If $k'^* < k^*$, then $k'^* < Q$ and

$$\Delta'(k'^*) < \kappa'_{k'^*} \leq \kappa_{k'^*} \leq \Delta(k'^*) \leq \Delta'(k'^*),$$

so $k^* \leq k'^*$. By M1 and M2, $\Delta_V(j) = V \Delta_1(j)$ with $\Delta_1(j) \geq 0$, and for u non-decreasing,

$$V \leq V' \Rightarrow \Delta_V \leq \Delta_{V'}, \quad c \leq c' \Rightarrow u(w) - u(w - c) \leq u(w) - u(w - c').$$

□

Proof of M12

Lean:

`EntryContestSpread.enters_down, EntryContestSpread.count_ge_of_enters, EntryContestSpread.count_le_of_fails, EntryContestSpread.margin_rise, EntryContestSpread.margin_fall.`

(i) The marginal entrant has $c < w_{k^*-1} \leq w'_{k^*-1}$, so

$$\kappa(w'_{k^*-1}) \leq \kappa(w_{k^*-1}) \leq \Delta(k^* - 1),$$

and for $i \leq k^* - 1$, with $w'_i \geq w'_{k^*-1} > c$,

$$\kappa(w'_i) \leq \kappa(w'_{k^*-1}) \leq \Delta(k^* - 1) \leq \Delta(i).$$

Every $i < k^*$ enters after the spread, so $k^* \leq k'^*$. (ii) For $k^* < Q$, rank k^* fails before the spread. If $w'_{k^*} > c$, then $c < w'_{k^*} \leq w_{k^*}$ and

$$\kappa(w'_{k^*}) \geq \kappa(w_{k^*}) > \Delta(k^*),$$

so rank k^* fails after the spread and $k'^* \leq k^*$. □

Proof of M13

Lean: `EntryContestSpread.pivot_rise, EntryContestSpread.pivot_fall, EntryContestSpread.spread_isPivot, EntryContestMM.mm2a.`

T is non-decreasing, so $w'_j = T(w_j)$ is sorted, and

$$x_0 \leq w_{k^*-1} \Rightarrow w_{k^*-1} \leq T(w_{k^*-1}), \quad w_{k^*} \leq w_{k^*-1} \leq x_0 \Rightarrow T(w_{k^*}) \leq w_{k^*},$$

so M12 applies. For $T(w) = x_0 + \lambda(w - x_0)$ with $\lambda \geq 1$,

$$T(w) - w = (\lambda - 1)(w - x_0), \quad \frac{1}{Q} \sum_j T(w_j) = \bar{w} + \lambda \left(\frac{1}{Q} \sum_j w_j - \bar{w} \right) = \bar{w} \quad (x_0 = \bar{w}).$$

□

Proof of M14

Lean: `EntryContestSpread.branch_of_not_band, EntryContestSpread.not_band_of_single_crossing, EntryContestSpread.pivot_single_crossing, EntryContestSpread.witness_counts, EntryContestSpread.witness_signs, EntryContestSpread.band_both_directions, EntryContestSpread.band_both_directions_positive.`

Outside the band $d_{k^*-1} \geq 0$ or $d_{k^*} \leq 0$, and M12 signs the change. If

$$i \leq j, d_j > 0 \Rightarrow d_i \geq 0,$$

then $d_{k^*} > 0$ gives $d_{k^*-1} \geq 0$. A pivot-spread has this property, since $d_j > 0$ gives $w_j > x_0$ and then $w_i \geq w_j > x_0$. With $c = 1$, $\kappa(w) = \frac{1}{w-1} - \frac{1}{w}$ and $\Delta(m) = \frac{1}{m+2}$, the profiles

$$w = (4, \frac{27}{10}, \frac{21}{10}), \quad w' = (4, \frac{53}{20}, \frac{13}{5}), \quad w'' = (4, \frac{9}{4}, \frac{11}{5})$$

have

$$\kappa(w) = (\frac{1}{12}, \frac{100}{459}, \frac{100}{231}), \quad \kappa(w') = (\frac{1}{12}, \frac{400}{1749}, \frac{25}{104}), \quad \kappa(w'') = (\frac{1}{12}, \frac{16}{45}, \frac{25}{66})$$

against $\Delta = (\frac{1}{2}, \frac{1}{3}, \frac{1}{4})$, so the counts are 2, 3 and 1, and d' and d'' both have signs $(0, -, +)$. Every inequality is strict, so by M9 the counts hold for every small $\mu > 0$. □

Proof of M15

Lean:

`EntryContestKappa.exit_below_pivot, EntryContestKappa.enters_downward, EntryContestKappa.p9_strict, EntryContestKappa.p9_monotone, EntryContestKappa.p9_strict_model.`

Let j be an entrant at $\lambda = 1$ with $w_j < x_0$. Since $\kappa(v) \rightarrow \infty$ as $v \downarrow c$, some $u > c$ has $\kappa > \Delta(j)$ on (c, u) , and

$$\lambda > \bar{\lambda} = \max\left(1, \frac{x_0 - u}{x_0 - w_j}\right) \Rightarrow x_0 + \lambda(w_j - x_0) < u,$$

so rank j fails at λ . By the prefix property no rank from j on enters, and

$$k^*(\lambda) \leq j < j+1 \leq k^*(1).$$

For $1 \leq \lambda_1 \leq \lambda_2$, ranks with $w_i \geq x_0$ have $i < j$ and enter at λ_1 , while ranks with $w_i < x_0$ have $x_0 + \lambda_2(w_i - x_0) \leq x_0 + \lambda_1(w_i - x_0)$, so every entrant at λ_2 enters at λ_1 . □

B Evidence and verification

C Terminology

D Primitives and scope

E Refuted conjectures

F Responses to the referee reports

References